

Theory Book

COMPUTER APPLICATIONS TECHNOLOGY

10



basic education

Department:
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REPUBLIC OF SOUTH AFRICA



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Computer Applications Technology Grade 10 Theory Book

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CONCEPTS OF COMPUTING

CHAPTER OVERVIEW

Unit 1.1	Information and communication technology
Unit 1.2	Data and information
Unit 1.3	Computers and ICT
Unit 1.4	Types of computers
Unit 1.5	Economic benefits of computers

At the end of this chapter, you should be able to:

- Understand ICT systems and how they are used in everyday life.
- Distinguish between data and information.
- Give examples of data and information used.
- Identify the main concepts of a computer.
- Understand the difference between hardware and software.
- Describe the information-processing cycle with reference to how it uses hardware devices in each stage.
- Describe the different types of computers and their uses.
- Explain the economic reasons for using computers.
- Understand the different social implications regarding the use of ICT.

INTRODUCTION

We use many different types of computers, or computer-controlled devices every day. This can range from smartphones and calculators, to the scanners used in supermarkets.

The history of computing goes back over 2 500 years to the abacus, which is a simple calculator made from beads and wires, and is still used in some parts of the world today.

In the following units, you will learn more about the different concepts used in computing, the different types of **ICT** systems used, as well as what the difference is between data and information. You will also look at the various components of a computer and the different types of computers available. Lastly, you will learn about the economic value of computers.

1.1 Information and communication technology

ICT SYSTEMS USED IN EVERYDAY LIFE

ICT systems are often confused with computers; however, they are not the same thing. Computers refer to the hardware that forms part of an ICT system. The ICT system refers to the overall set-up, consisting of hardware, **software**, data and the people that use it. It commonly includes communication technology, such as the internet.

Information and communication technology (ICT) is a field that is related to computers and communication networks. It can also be in the form of audiovisuals, for example, film and digital productions. Digital technologies use data, such as text, numbers, images, sounds and videos. Computers use this data to show information.

In the past, these technologies used to be separate systems, but recently, these technologies have become almost “one”. Computers have become more like smartphones and smartphones have become more like computers. ICT is the field that looks at all these technologies as one.

An ICT system depends on computers, data and people to make the system work. For ICT systems to function correctly, they need to receive, store, retrieve, manipulate and transmit data. Data is often in the form of text, numbers, sounds and images.

EXAMPLE OF AN ICT SYSTEM IN EVERYDAY LIFE

Computers, however, cannot act on their own; a computer needs someone to give it (enter) the data that needs to be processed. To understand how it works, look at the following example.

Supermarkets and chain stores use ICT systems for a variety of reasons. In this example, we will look at the processes of receiving, manipulating, storing and communicating data at a point-of-sale (POS) system.

PROCESSES AT A POS SYSTEM

When you look at a supermarket's ICT system from the cashier and customer's point of view, all you see is a stand-alone computer. This computer is called a cash register. However, the cash register consists of much more than that.

All items have a **barcode**, which is a pattern of narrow and wide stripes that can be read by a barcode reader. The barcode reader is used to scan the barcode optically and convert the stripes into numerical data, which can automatically be transferred to the check-out computer (i.e. the cash register). The barcode reader sometimes beeps after scanning an item to signal to the cashier that the item has been identified.



Figure 1.1: An example of a barcode

INPUT AND STORING DATA

When an item is scanned, the check-out computer checks for the price of the item in the database. The running total of each item is stored temporarily in the check-out terminal. (There may be other data stored there as well, such as money that was taken out of the till during the day.)

MANIPULATING DATA

After all the items have been scanned, the check-out computer calculates the data to determine the total cost. If the customer pays with cash and requires change, the cashier will enter the amount given and the check-out computer will calculate any change that should be given to the customer. This is an example of the computer receiving data from the user.

TRANSMITTING DATA

When all the items are scanned into the check-out computer, the customer can see information about the price of each item, which is shown on a small display.

After all the items have been scanned, the total amount is shown. The customer then pays and is given a printed till receipt showing all the items that were bought and how much each item costs.



Figure 1.2: *Cash register and receipt*

COMMUNICATION

In some instances, a product's data can also be sent to the shop's main computer for stocktaking purposes, as well as to the manufacturer for their stock purposes.



Activity 1.1

1. Explain the term ICT. Do not just expand on the acronym.
2. What does POS stand for?
3. The following table shows a few activities that take place before check-out. Match these activities with one of the following processes associated with a POS, as seen from the customer's point of view:
 - Receives
 - Sends
 - Stores
 - Retrieves
 - Manipulates

ACTIVITY	PROCESS
Scan a barcode	
Calculate change	
Provide a receipt	
Total cost of items purchased	

4. Imagine a world without ICT systems, or ICT-controlled devices. ICTs have several advantages and disadvantages. Copy and fill in the table below, and list three advantages and disadvantages of ICT systems.

ADVANTAGES	DISADVANTAGES

1.2 Data and information

In computing, it is very important to understand the difference between data and information. This section will look at the differences between the two and the importance thereof.

What do you think of first when you hear the word *data*? What comes to mind is raw material, which refers to something that has not yet been processed. Data needs to be processed before it can become useful. Data can be in the form of text, words, numbers or symbols which, if not put into some type of context, is pretty useless to us.

Information, on the other hand, is data that is processed and formatted in such a way that we can actually use it. Both data and information are important; it is through accurate data collection that people, such as managers, have the necessary information to make informed decisions. Let's take a look at the following examples of data and information:

Example of data	5605, David, 30, Bisho, 0731112222, Ramaphosa, Mahlangu, Avenue
Example of information	David Mahlangu 30 Ramaphosa Avenue Bisho 5605 (073) 111 2222

As you can see from this example, the data appears to be a set of random words and numbers. Only after the data has been interpreted, organised and formatted, you can see that it is the contact details for David Mahlangu.

Let's look at the following example of how data can be converted into useful information.

DETERMINING HOW THE LEARNERS DO IN MATHEMATICS

Mandla's Mathematics teacher needs to determine how the learners in her class are performing. She must get the test results (data) and the average of each learner, as well as the average mark for the entire class (information).

There are five learners in the Mathematics class and so far, they have written three tests.

Table 1.1: Data about learners' test results for three tests

				
James	Mandla	Lucy	Jarred	Zanele
$\frac{13}{20}$ $\frac{20}{30}$ $\frac{7}{15}$	$\frac{14}{20}$ $\frac{18}{30}$ $\frac{11}{15}$	$\frac{10}{20}$ $\frac{23}{30}$ $\frac{9}{15}$	$\frac{13}{20}$ $\frac{21}{30}$ $\frac{8}{15}$	$\frac{18}{20}$ $\frac{25}{30}$ $\frac{8}{15}$

... continued



Animation

Make an animation similar to what is shown in the following video:

<https://www.youtube.com/watch?v=Rs57-PQyqaE>

You can also use baking a cake as example. This example will include the following as data: flour, baking soda, salt, milk, eggs, icing sugar, the oven as the computer and the cake as information.

DETERMINING HOW THE LEARNERS DO IN MATHEMATICS

Table 1.2: Maths results Term 1 (sorted from highest to lowest score in percentage)

	TEST 1	TEST 2	TEST 3	TOTAL	AVERAGE (%)
Learner	20	30	15	65	
Zanele	18	25	8	51	78
Mandla	14	18	11	43	66
Lucy	10	23	9	42	65
Jarred	13	21	8	42	65
James	13	20	7	40	62
TERM AVERAGE MARK					67

Tables 1.1 and 1.2 show the difference between data and information. In Table 1.1, we cannot see how the learners are performing. However, after the computer manipulated the test scores, we are given meaningful information, as can be seen in Table 1.2. Remember, the information we get out of a computer is only as good as the data that is entered into it.

THE GIGO PRINCIPLE

GIGO is an acronym for *Garbage In, Garbage Out*. Simply put, it means that bad input will result in bad output. It is very important to remember the GIGO principle, because bad data will give you bad information.

Using the example of the learners' test results, if the teacher entered the incorrect marks into the computer (i.e. bad data), the information that she would get as a result, will also be incorrect.

It is quite possible for data to be interpreted incorrectly. This is often the result of incorrect or incomplete data, or a lack of context.

THE INFORMATION-PROCESSING CYCLE

For you to understand how a computer works, you need to understand the information-processing cycle. The stages in this cycle repeat itself over and over again and is made up of the following:

- **Input:** In the first stage of computing, the computer receives some data or instructions to follow.
- **Processing:** In the second stage of computing, the computer follows the instructions programmed into it and manipulates the data in some way.
- **Output:** In the third stage of computing, the computer supplies the new, manipulated information in some way. This information can be displayed on the screen (for example, the message you see on the screen while you type), or it can be sent to a different part of the computer where it is received as input.
- **Storage:** An **optional** stage is where data is stored. The data can be stored permanently (on a **hard drive**), or temporarily (on RAM – i.e. the computer's short-term memory).

- **Communication:** Another optional stage is communication, where the output created is sent across a network to other computers. While this step used to be rare in the past, most computer programs perform some type of communication today.

An example of the information-processing cycle is shown in the following figure:

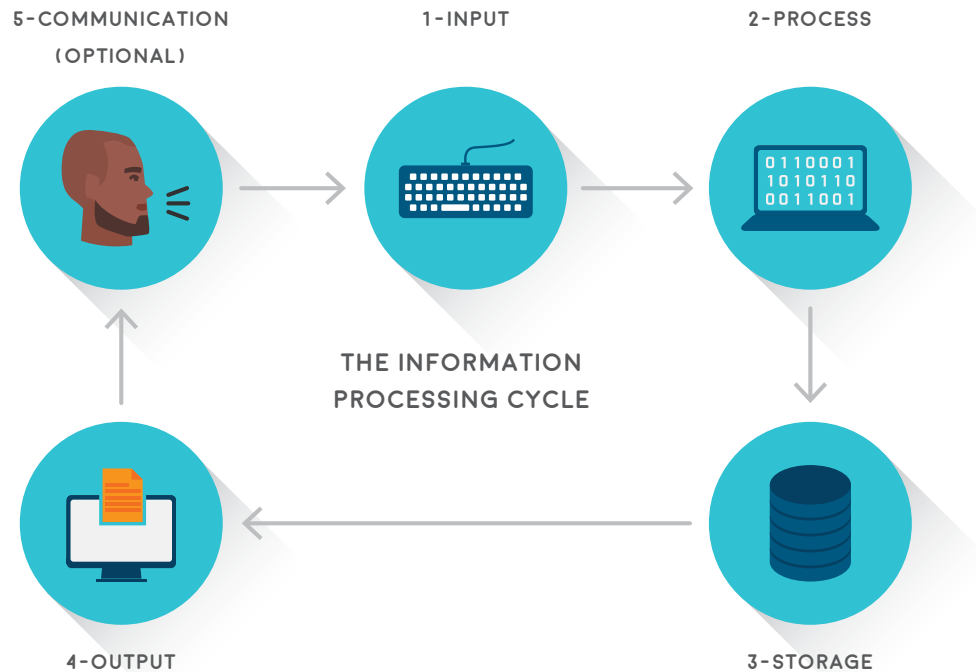


Figure 1.3: The information-processing cycle

The information-processing cycle is a series of stages carried out to get information from raw data. Although each of the first three stages explained above (input, processing and output) must be taken in order, the order is **cyclic**. The output stage can lead to the repeat of the data-collection stage, resulting in another cycle of data processing.

Activity 1.2

1. Sizwe is in Grade 10 and received his school report. When he looks at his report, this is what he sees:

SUBJECT	AVERAGE 73
Life Orientation Physical Science	80
Mathematics	67
Computer Applications Technology	65
Geography Business Studies	73 88 67

- 1.1 What is wrong with this report?
- 1.2 Which principle is referred to above?
- 1.3 Explain the difference between data and information.

... continued



Activity 1.2

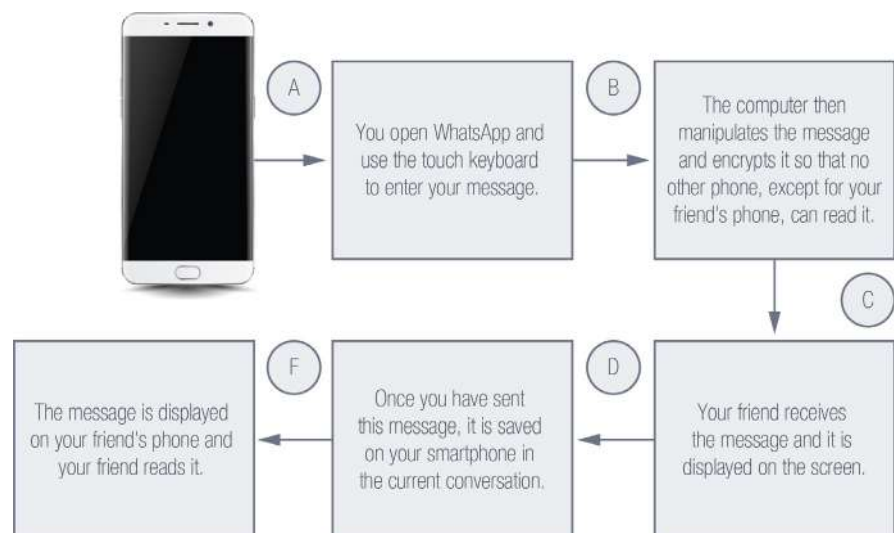
... continued

- 1.4 Use the data from the previous page and create a report showing the correct information for Sizwe. Use the table below to guide you:

SUBJECT	TERM MARK (%)
Average	

2. Answer the following questions based on the scenario below.

Computers are electronic devices that manipulate data and turn it into useful information. To see how this works, think about sending a WhatsApp message from a cell phone to a friend. The diagram below shows the different stages in sending a WhatsApp message.



- 2.1 What process is shown in this diagram?
- 2.2 List the different stages from A to F.
- 2.3 What happens during Stage C of this cycle?
- 2.4 Indicate whether the following is input, output, or storage:
- a. Typing a message to a friend on WhatsApp
 - b. Music that you listen to from your phone
 - c. Printing a file
 - d. Playing videos using your phone
 - e. Photos that you keep on your phone

1.3 Computers and ICT

Computers are electronic devices that receive data, manipulate the data and then turn it into useful information, before storing it and/or sending it to someone, or somewhere, to be read or acted upon.

The following figure tries to explain how a computer works. You can see that the functions of a computer are similar to the stages of the information-processing cycle.

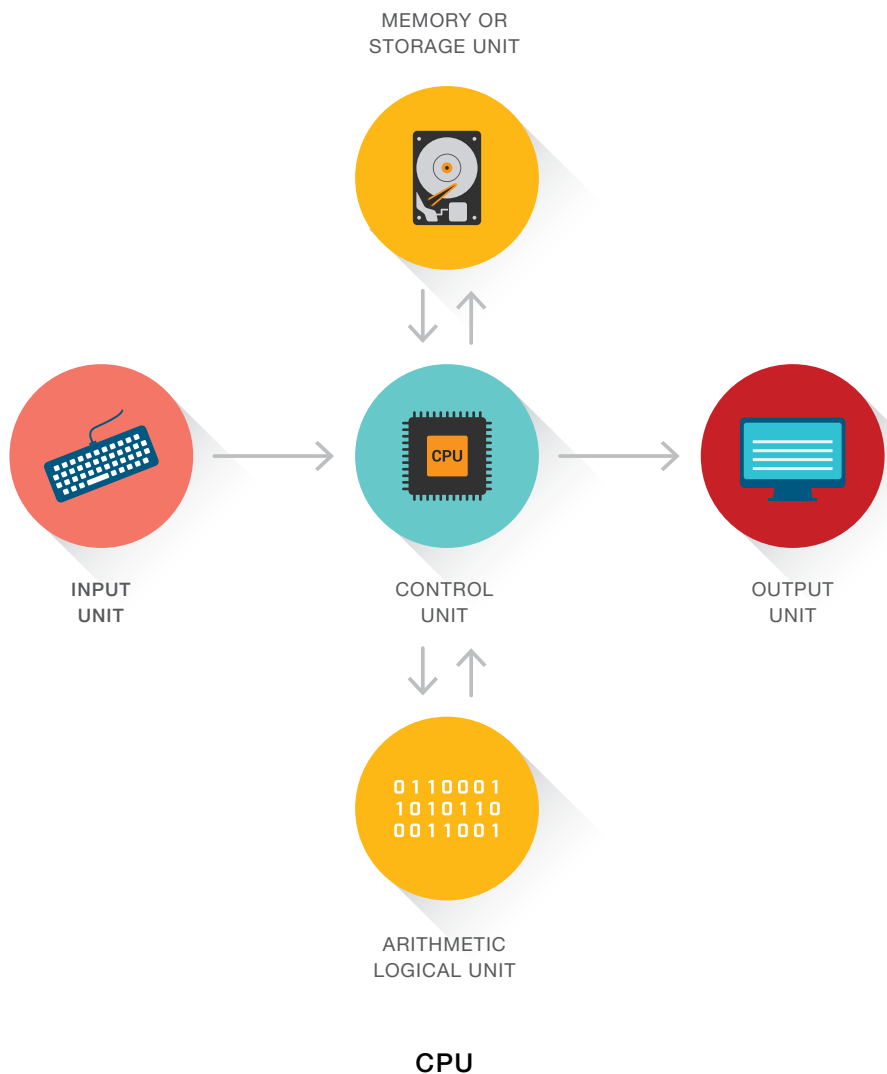


Figure 1.4: General model of a computer

Looking at the figure, you will see that all basic computers consist of four functions:

1. Input
2. Storage
3. Processing
4. Output

These four things are combined to make a computer work.

For a computer to function properly, it must consist of both hardware and software, which both depend on each other. Without software, the hardware of a computer will have no



WHAT MAKES A
COMPUTER A
COMPUTER?



<https://www.youtube.com/watch?v=mCq8-xTH7jA>

function and vice versa. Both of these concepts are discussed in detail in the chapters to follow; however, we will just briefly explain what each means.

- **Hardware:** The physical components of a computer, also referred to as the *equipment* of a computer.
- **Software:** More commonly known as apps (applications) or programs and consists of a list of instructions in a computer language that instructs the computer on what to do.

Examples of hardware devices consist of the following:

- Keyboard and mouse used for input
- Monitors, printers and speakers used for output
- Hard drives and flash disks used for storage
- Routers used for communication

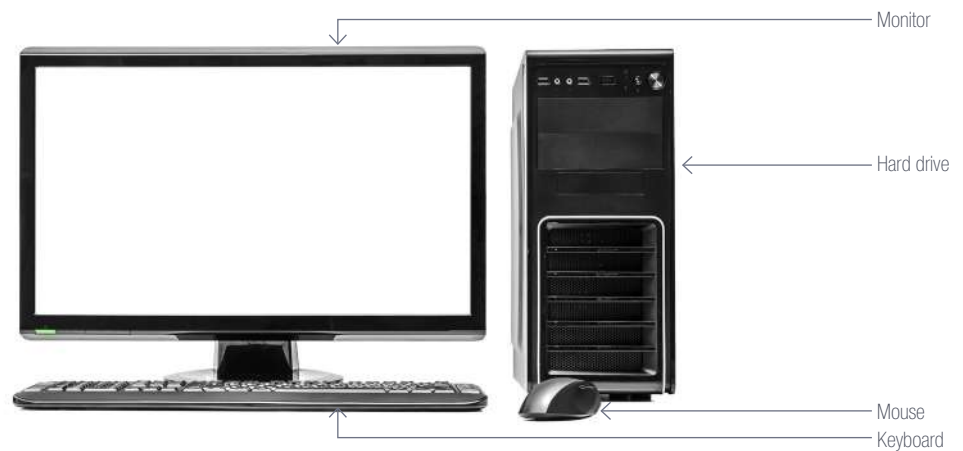


Figure 1.5: Computer hardware



Activity 1.3

1. What is a computer?
2. Match each concept in Column A with the correct definition in Column B. Write only the number and the letter.

COLUMN A	COLUMN B
1. Hard disk	A. Can be text, numbers and figures that are used by computers
2. Data	B. Storage device
3. Input	C. Bad input will result in bad output
4. GIGO	D. The components that make up the computer
5. Output	E. The computer receives instructions to follow
6. Hardware	F. The information produced by a computer

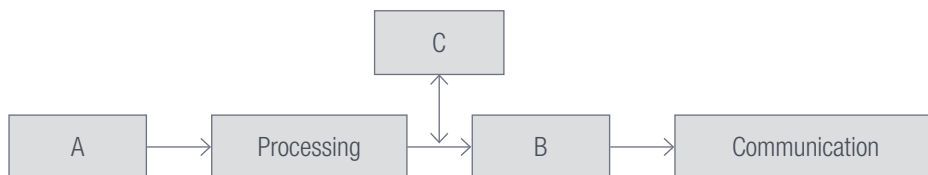
... continued



Activity 1.3

... continued

3. Answer the questions based on the diagram below.



- 3.1 Name the missing stages, as indicated by the letters below:
- A:
 - B:
 - C:
- 3.2 List one example of each of the following:
- Hardware
 - Software
- 3.3 Your teacher is busy printing out the term class reports. Which stage is this in the information-processing cycle?
4. Complete the following table and identify two hardware devices that your teacher will use for each of the missing stages. Motivate why she would use those specific devices.

STAGE	HARDWARE DEVICE	REASON
A		
B		
C		

1.4 Types of computers

As you know, computers are devices used to process information. There are several terms used to describe computers; most of these terms indicate the size and the capability of the computer. The two main categories of computers are:

1. **Multi-purpose:** These types of computers have an operating system that can run many applications at the same time. Some examples are devices, such as servers, laptops and smartphones.
2. **Dedicated devices:** These are devices that can only run one task. Some examples are ATMs and processors embedded in devices, such as refrigerators and motor cars.



Figure 1.6: Different types of computers

MULTI-PURPOSE DEVICES

Table 1.3: Different types of general-purpose computers

DESCRIPTION	EXAMPLES OF WHERE THIS TECHNOLOGY IS USED
Supercomputer  A supercomputer is the fastest, most powerful and most expensive type of computer made today. It is used to perform many instructions per minute.	Supercomputers are used for difficult or complicated calculations, such as: • Predicting the weather • Locating possible oil deposits

... continued